AMPLE ON THE REDUCTION AFTER PASSING TO THE LIMIT

THE STACKS PROJECT AUTHORS

Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Noetherian ample-on-reduction. Cohomology of Schemes, ample on reduction lemma
- Limit characterization of ampleness. Limits, ampleness lemmas of the chapter

Source reference. Original reference 0892: sources/stacks-chow/tags.txt, source lines 10482–10482. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MP: sources/stacks-chow/tags.txt, source lines 12187–12187. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MQ: sources/stacks-chow/tags.txt, source lines 12188–12188. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZR: sources/stacks-chow/tags.txt, source lines 2604–2604. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 056A: sources/stacks-chow/tags.txt, source lines 6703–6703. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09MT: sources/stacks-chow/tags.txt, source lines 12191–12191. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05F4: sources/stacks-chow/tags.txt, source lines 7013–7013. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09MS: sources/stacks-chow/tags.txt, source lines 12190–12190. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes inducing a homeomorphism of underlying topological spaces. Let $\mathcal{L}$ be an invertible sheaf on X. Then $i^*\mathcal{L}$ is ample on Z, if and only if $\mathcal{L}$ is ample on X.*

Proof. If $\mathcal{L}$ is ample, then $i^*\mathcal{L}$ is ample for example by Morphisms, Lemma 0892. Assume $i^*\mathcal{L}$ is ample. Then Z is quasi-compact (Properties, Definition 01PS) and separated (Properties, Lemma 09MP). Since i is surjective, we see that X is quasi-compact. Since i is universally closed and surjective, we see that X is separated (Morphisms, Lemma 09MQ).

By Proposition 01ZA we can write $X = \lim X_i$ as a directed limit of finite type schemes over $\mathbf{Z}$ with affine transition morphisms. We can find an i and an invertible sheaf $\mathcal{L}_i$ on X_i whose pullback to X is isomorphic to $\mathcal{L}$, see Lemma 01ZR.

For each i let $Z_i \subset X_i$ be the scheme theoretic image of the morphism $Z \to X_i$. If $\mathrm{Spec}(A_i) \subset X_i$ is an affine open subscheme with inverse image $\mathrm{Spec}(A)$ in X and if $Z \cap \mathrm{Spec}(A)$ is defined by the ideal $I \subset A$, then $Z_i \cap \mathrm{Spec}(A_i)$ is defined by the ideal $I_i \subset A_i$ which is the inverse image of I in A_i under the ring map $A_i \to A$, see Morphisms, Example 056A. Since $\mathrm{colim}\, A_i/I_i = A/I$ it follows that $\lim Z_i = Z$. By Lemma 09MT we see that $\mathcal{L}_i|_{Z_i}$ is ample for some i. Since Z and hence X maps into Z_i set theoretically, we see that $X_{i'} \to X_i$ maps into Z_i set theoretically for some $i' \geq i$, see Lemma 05F4. (Observe that since X_i is Noetherian, every closed subset of X_i is constructible.) Let $T \subset X_{i'}$ be the scheme theoretic inverse image of Z_i in $X_{i'}$. Observe that $\mathcal{L}_{i'}|_T$ is the pullback of $\mathcal{L}_i|_{Z_i}$ and hence ample by Morphisms, Lemma 0892 and the fact that $T \to Z_i$ is an affine morphism. Thus we see that $\mathcal{L}_{i'}$ is ample on $X_{i'}$ by Cohomology of Schemes, Lemma 09MS. Pulling back to X (using the same lemma as above) we find that $\mathcal{L}$ is ample. □